Closing the center-embedding gap in Spanish: can grammatical features do the work of memory?

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Abstract

Center embeddings (clauses nested inside one another in the left branches of the structure) challenge comprehension. They are a natural probe of the memory architecture underpinning language (Karlsson 2007) Their hallmark is the missing-VP illusion: when one of the three necessary verbs is omitted (examples below), readers perceive the ungrammatical sentence as more acceptable than the grammatical one, as if a retained prediction has subtly deteriorated (Lewis and Vasishth 2005; Hahn et al. 2022). The illusion has been demonstrated recently in Spanish Pañeda and Lago 2024. Nonetheless, the degree of the illusion was aligned with the statistical scarcity of the verb sequence, and individual working memory span provided no predictive insights. But storage capacity alone does not seem to be the right variable.

The Spanish language contains relativizers with distinct morphological attributes but consistent syntactic placements: the unmarked *que* compared to *el cual* and *quien*, which reflect gender, number, and animacy. If the illusion is a result of the ambiguous reconstruction of compromised memory traces, then feature-bearing relativizers should furnish distinctive retrieval cues and mitigate the illusion, without modifying sentence length, structure, or embedding depth. Featureless *que* can not.

We will evaluate this with 120 participants employing speeded acceptability judgments, eye-tracking, and EEG, anticipating a gradual decrease in illusion frequency correlated with feature richness, a diminished sustained anterior negativity during retention, and a reinstated P600 at the absent verb.

The result would be a linguistically precise manipulation that would shift the focus of the brain's ability to maintain structure open to grammatical features, rather than capacity. This would also raise a generally neurobiological question: what distinguishes a memory cue? **Keywords:** sentence processing; cue-based retrieval; working memory; morphosyntactic features; ERP; Spanish

SUPPORTING MATERIALS

The critical manipulation, on one item. All conditions are double center embeddings of identical depth; the featural content of the relativizer is crossed with grammaticality (VP2 entrevistó present vs. deleted).

(1) a. La política **a la que** el periodista **al que** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a la que** el periodista **al que** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
BASELINE — *que*, gender/number only via the article; the condition used in all prior work.

(2) a. La política **a la cual** el periodista **al cual** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a la cual** el periodista **al cual** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
FEATURE-BEARING — *el cual* inflects for gender and number on the pronoun itself.

(3) a. La política **a quien** el periodista **a quien** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a quien** el periodista **a quien** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
ANIMACY-BEARING — *quien* restricts the antecedent to a human referent.

Differential object marking on *a* blocks a subject-relative reading throughout, so no condition is ambiguous at the relativizer.

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